ABDULLAH AL KHALID

Software Engineer / Distributed Systems

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SUMMARY

Software engineering graduate focused on distributed systems, databases, and backend infrastructure. Experience with .NET APIs, SQL data pipelines, and systems work in Rust and Go — including Raft, gossip protocols, CRDTs, and append-only logs.

WORK EXPERIENCE

DKSH **Kuala Lumpur, Malaysia**

Software Engineer Intern Sep 2024 – Feb 2025

Worked with cross functional teams to keep production applications running smoothly, building a website plugin, automating manual work, and migrating SQL Server data through .NET Core APIs and pipelines.

EDUCATION

Universiti Teknologi Malaysia (UTM) **Skudai, Johor, Malaysia**

Bachelor of Computer Science with Honours in Software Engineering 2021 – 2025

PASSION PROJECTS

ironbarrel (Rust)

A log-structured hash table key-value storage engine inspired by the orignal bitcask

- Built a high throughput storage engine from ground up in Rust, inspired by the Basho Bitcask paper; designed lock-free concurrent indexing (KeyDir) and positional reads (pread) to achieve 280k+ ops/sec with 1.2µs p50 read latency.
- Engineered a crash-resilient storage & compaction pipeline using hardware-accelerated CRC32 and binary hint-file indexing, cutting 1M-key startup recovery by 96% (from 2.4s to 80ms) while automatically reclaiming dead disk space.

Miserable Messengers / Gossip Gloomers (Rust, Maelstrom/Jepsen)

Fly.io distributed systems challenge series

- Built a Rust node engine that solved Fly.io distributed systems challenges, including gossip broadcast, CRDT counters, and a linearizable KV store.
- Verified correctness under network partitions, packet drops, and concurrent clients using Jepsen's Maelstrom;

ACTIVITIES & COMPETITIONS

RecyclO **Project**

Plastic Waste Management System

Designed a plastic waste management system aimed at reducing plastic waste through practical collection and processing workflows. Focused on collaborative, sustainable solutions intended to reduce environmental impact.

Flood Remission System **Project**

Disaster Management

Developed a multipurpose disaster-management application focused on flood mitigation. Used real-time weather sensors and satellite imagery to track flood conditions and forecast flooding events. Sent early-warning alerts to residents in flood-prone areas and supported authorities in coordinating rescue, relief, and shelter.

REFERENCES

A. S. M. Khairuzzaman
Managing Director & CEO (Acting); Deputy Managing Director & COO
Central Counterparty Bangladesh PLC
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